Inventory Management System

Description

The **Inventory System Management** project is a comprehensive desktop application designed to help businesses efficiently manage their inventory, suppliers, employees, and sales. Built using **Python** with a **Tkinter** graphical user interface, this system provides a user-friendly platform for organizing and tracking stock items while automating routine tasks.

Key highlights of the system include:

- **User Authentication:** Secure login for authorized access to the system.
- **Product Management:** Add, edit, delete, and search products with detailed information such as category, supplier, and stock levels.
- **Category & Supplier Management:** Easily manage product categories and supplier details for streamlined inventory organization.
- **Employee Management:** Maintain records of employees who manage and operate the inventory system.
- **Sales Management:** Record sales transactions and generate reports to track performance and stock usage.
- **Dashboard:** A visually interactive dashboard displaying real-time statistics about inventory, sales, and other key metrics.

This system is ideal for small to medium-sized businesses looking to reduce manual record-keeping, improve stock monitoring, and make data-driven decisions for inventory management.

Features

The **Inventory System Management** project includes the following key features:

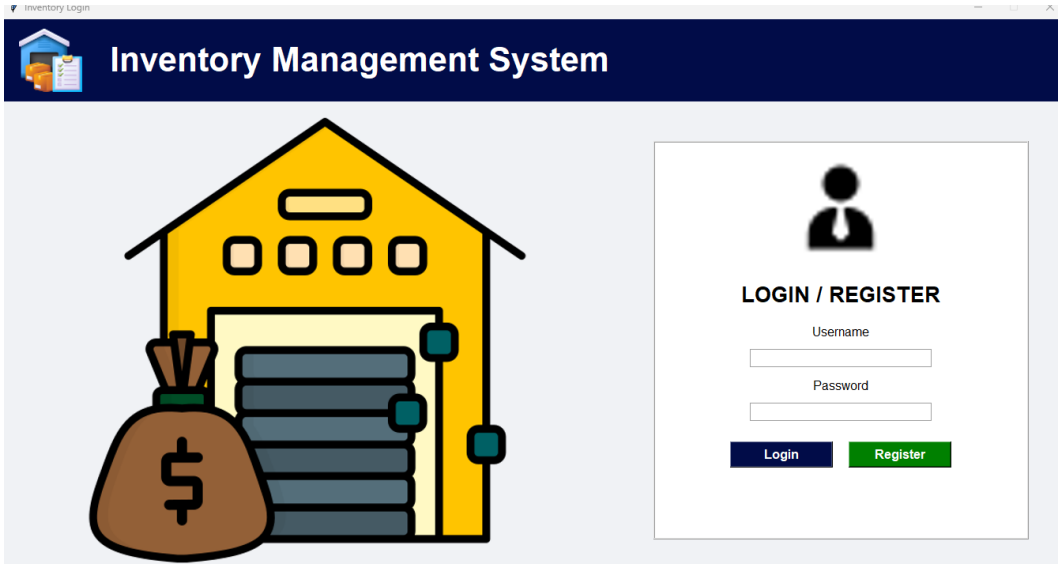
1. **User Login & Authentication**
   - Secure login system to ensure only authorized users can access the application.
   - Protects sensitive inventory and sales data.
2. **Product Management**
   - Add new products with details such as name, category, supplier, price, and stock quantity.
   - Edit and update existing product information.
   - Delete products that are no longer in inventory.
   - Search and filter products quickly.
3. **Category Management**
   - Create, edit, and delete product categories for better organization.
   - Categorize products to easily manage stock.
4. **Supplier Management**
   - Maintain supplier details including name, contact, and address.
   - Add, update, or remove suppliers as needed.
5. **Employee Management**

    - Store and manage employee information.
    - Track which employees handle inventory or sales tasks.
6. **Sales Management**
    - Record sales transactions with product details, quantity, and total price.
    - Automatically update product stock after each sale.
    - Generate sales reports for monitoring performance.
7. **Dashboard & Analytics**
    - Visual dashboard displaying key metrics like total products, sales, and employees.
    - Provides a quick overview of business performance.
8. **User-Friendly Interface**
    - Intuitive **Tkinter GUI** for easy navigation.
    - Buttons and forms for performing actions quickly and efficiently.
9. **Data Persistence**
    - All data is stored in a **MySQL database**, ensuring secure storage and easy retrieval.
10. **Error Handling & Alerts**
    - Pop-up messages for successful operations, warnings, and errors.
    - Prevents accidental data loss or incorrect entries.

Technology Stack

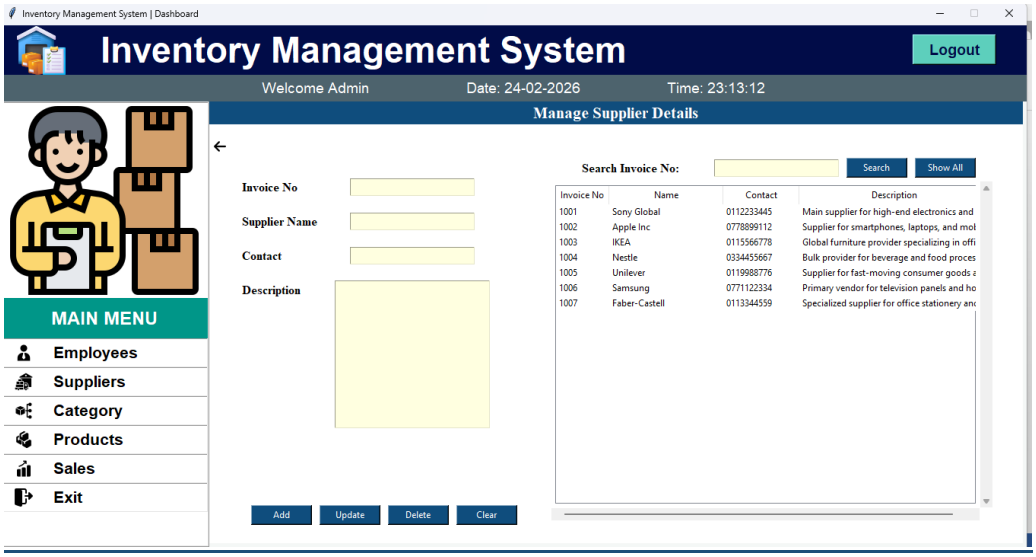
- Example:
    - Python 3.x
    - Tkinter (GUI)
    - MySQL / pymysql (Database)
    - PIL (for images)
    - Tkcalendar

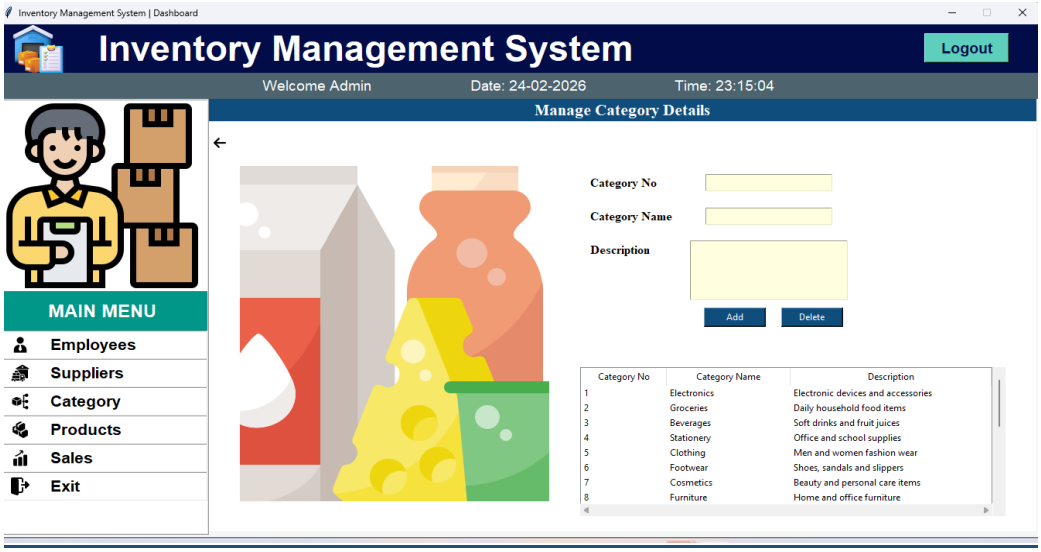
Usage

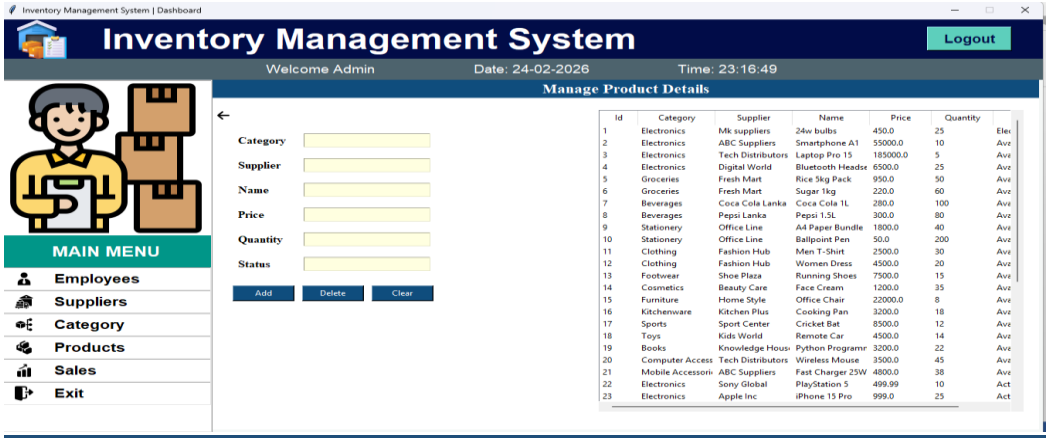


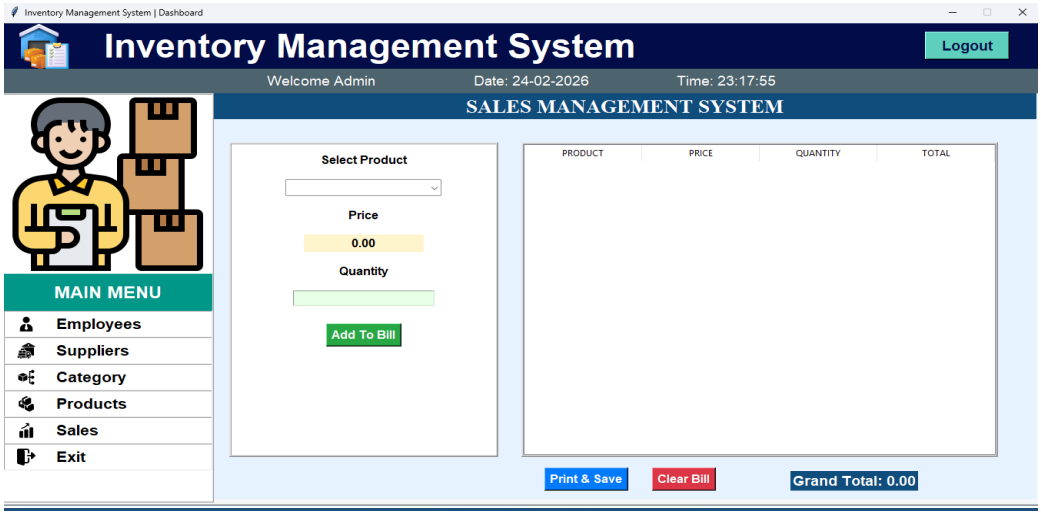
Inventory Management System
Logout
Welcome Admin
Date: 24-02-2026
Time: 23:09:26
MAIN MENU
Employees
Suppliers
Category
Products
Sales
Exit
Total Employees
5
Total Suppliers
7
Total Categories
20
Total Products
39
Total Sales
3

Inventory Management System | Dashboard
Inventory Management System
Logout
Welcome Admin
Date: 24-02-2026
Time: 23:12:22
Manage Employee Details
Search By
Search
Show All
Employee | Name | user_email | Gender | Date of Birth | Contact | Type of Employment | Education | Work Shift
1 | Deleepa Srimal | dulanjalee@gmail.com | Male | 21/03/00 | 715979672 | Casual | M.Tech | Evening
2 | Dulanjalee Priyanga | deleepa@gmail.com | Female | 21/03/00 | 076-4525504 | Casual | B.Sc | Evening
3 | Shamila Bawanthi | shamila@gmail.com | Female | 11/02/16 | 079-4562874 | Contract | M.Tech | Evening
6 | Merian Mallika | mallika@gmail.com | Female | 10/02/06 | 078-5969321 | Casual | M.Com | Evening
7 | Imangi Perera | imangi@gmail.com | Male | 13/02/03 | 078-5965896 | Casual | BBA | Morning
MAIN MENU
Employees
Suppliers
Category
Products
Sales
Exit
Employee ID
Name
Email
Gender
Select Gender
Date of Birth
24/02/26
Contact
Employment Type
Select Type
Education
Select Education
Work Shift
Select Shift
City
Date of Joining
24/02/26
Salary
User Type
Select Usertype
Password
ADD
Update
Delete
Clear


Inventory Management System | Dashboard
Inventory Management System
Logout
Welcome Admin
Date: 24-02-2026
Time: 23:13:12
Manage Supplier Details
Invoice No
Supplier Name
Contact
Description
Search Invoice No:
Search
Show All
Invoice No | Name | Contact | Description
1001 | Sony Global | 0112233445 | Main supplier for high-end electronics and
1002 | Apple Inc | 0778899112 | Supplier for smartphones, laptops, and mob
1003 | IKEA | 0115566778 | Global furniture provider specializing in offi
1004 | Nestle | 0334455667 | Bulk provider for beverage and food proces
1005 | Unilever | 0119988776 | Supplier for fast-moving consumer goods a
1006 | Samsung | 0771122334 | Primary vendor for television panels and ho
1007 | Faber-Castell | 0113344559 | Specialized supplier for office stationery and
MAIN MENU
Employees
Suppliers
Category
Products
Sales
Exit
Add
Update
Delete
Clear







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